



The Floor Effect on the ABAS-II in Adolescents with Autism Spectrum Disorder

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Abstract

The present study aimed at enhancing the assessment of adaptive behavior in adolescents with Autism Spectrum Disorder through the Adaptive Behavior Assessment System-Second Edition. To overcome the limits underpinning floor effect, which are commonly observed with the traditional methods, the statistical Hessel method and its revision by Orsini, Pezzuti and Hulbert was adopted. The results showed a more variation using the former method compared to the latter: a greater impairment in Social versus Leisure skills emerged. Regarding the Conceptual domain, a greater deficit in Functional Academics was observed. In the Practical domain, we found a greater impairment in Self-Care skills. Conversely, Home Living skills appeared as a strength compared to other skills. The results are discussed in reference to literature.

Keywords Autism spectrum disorder · Floor effect · Adaptive behavior · ABAS-II · Adolescence

Introduction

Adaptive behavior is a set of skills that allows people to meet age-related societal expectations and standards of personal independence and social responsibility. Specifically, adaptive behavior encompasses conceptual, social and practical skills that are necessary to perform independently

during daily activities across different settings, as at home, at school/work and within the community (Schalock et al., 2010; Sparrow et al., 2005; Tassé & Grover, 2013). These skills are learned throughout development, and they become more advanced with age (Harrison, 1990; Widaman et al., 1991).

Interest in adaptive behavior began growing at the end of 1950s, when it was first included among criteria for intellectual disability (Heber, 1961). While it remains fundamental in the diagnosis of the severity of intellectual disability (APA, 2013), impairments in adaptive behavior may also be observed in other neurodevelopmental disorders, as Autism Spectrum Disorder (ASD) and Attention Deficit Hyperactivity Disorder (ADHD), as well as in other behavioral, emotional and learning difficulties with partially intact cognitive functions (Ditterline & Oakland, 2009; Tasse et al., 2012).

The assessment of adaptive behavior appears very complex for a variety of reasons. First, it must evaluate level, quality, fluency and pattern across different domains (e.g., communication, socialization, school, free time, motor), in order to produce a comprehensive report of strengths and weaknesses. It is important to distinguish what a person can do from what they really do in everyday living situations. Second, the direct observation of adaptive behavior is seldom practical, considering one would have to observe a person in a wide variety of settings attempting many different skills. Thus, indirect methods are usually preferred

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and include interviews—unstructured, semi-structured or structured—and checklists. These measures are administered or completed by parents, teachers or other caregivers who live in close contact with the person being evaluated (Ferri et al., 2014; Sparrow et al., 1984). Third, it is not unusual to observe a floor effect—the lowest level of behavioral performance assessed—when raw scores are converted into standard scores. This effect may determine a flat profile, thus obscuring some strengths (Fenton et al., 2003).

In Italy, only three instruments are norm-referenced scores, have good psychometric proprieties, and allow to evaluate multiple domains of adaptive behavior: The Adaptive Behavior Assessment System-Second Edition (ABAS-2; Harrison & Oakland, 2003; Ferri et al., 2014), the Vineland Adaptive Behavior Scales-Second Edition (VABS-II; Balboni et al., 2016; Sparrow et al., 2005), and the Adaptive Behavior Inventory (ABI; Brown & Dario, 1994; Brown & Leigh, 1986). In a study comparing adaptive behavior measures for children with ASD (Lopata et al., 2013), scores on social domain derived from the ABAS-II and the VABS-II did not significantly differ each other. However, the ABAS-II showed a higher sensitivity compared to the VABS-II, capturing lower-level adaptive skills underpinning other domains (e.g., daily living skills, communication). For this reason, we focused on the ABAS for our study.

The ABAS allows a comprehensive evaluation of adaptive functioning in the home and community. It is available in different versions according to age: Parent/Primary Caregiver Form (0–5 years), Parent Form (5–21 years), Teacher/Day Care Provider Form (2–5 years), Teacher Form (5–21 years), Adult Form (16–89 years). The Parent Form—which was used in the current study—includes the assessment of ten skill areas (Communication, Community Use, Functional Academics, Home Living, Health and Safety, Leisure, Self-Care, Self-Direction, Social, and Work) and an overall adaptive functioning, namely the General Adaptive Composite. Ten skills are combined to form three broad domains of adaptive behavior: Conceptual, Social and Practical. The Conceptual domain includes Communication, Functional Academics, and Self-Direction. The Social domain encompasses Social and Leisure. The Practical domain includes Community Use, Home Living, Health and Safety, and Self-Care.

Autism Spectrum Disorder and Adaptive Behavior

ASD is a neurodevelopmental disorder whose prevalence is estimated to vary from 1.9/10,000 to 72.6/10,000 in Europe (for systematic review, see Elsabbagh et al., 2012), with a male-to-female ratio of 4:1 (Baio, 2014; Rappotti ISTISAN, 2013). ASD is characterized by deficits in social communication and social interactions as well as the presence of restricted and repetitive behaviors and interests. The

latter refers to stereotypical or repetitive motor movements, rigid routines, difficulties in coping with change, and narrow interests, whereas deficits in social communication and interactions include impairments in social-emotional reciprocity, non-verbal communication behaviors and interpersonal relationships (APA, 2013).

There is a wide variability in the diagnostic characteristics of people with ASD that concerns the type of symptoms experienced, the onset and duration of symptoms, and which areas of functioning are compromised (Volkmar et al., 2004; Wozniak et al., 2017). Furthermore, cognitive functioning contributes to variability within the group of individuals with ASD, with about 44% of children with ASD having a medium or medium-high intelligence quotient (IQ; Christensen et al., 2016; Pugliese et al., 2016). Beyond intelligence quotient and cognitive abilities, the variability of autism symptoms is also associated with linguistic and adaptive behavior, and this phenotypic variability tends to change over time (Billstedt et al., 2007; Georgiades et al., 2013). Caregivers report more severe symptoms of autism in early childhood than they do in adolescence and adulthood (Billstedt et al., 2007; Taylor & Selzer, 2010). Billstedt et al. (2007) suggest that adults with ASD are less likely to have serious communication problems (e.g., delayed echolalia) than younger people. However, in adolescence it is possible to observe marked regressions in cognitive and behavioral domains (Périsse et al., 2010).

The link between adaptive functioning and variables like age, IQ and ASD symptoms has not been fully understood perhaps due to mixed results provided by studies, where different measurements and sampling techniques were used (Lopata et al., 2012; McDonald et al., 2015, 2016). In general, positive relations between IQ and adaptive functioning, as evaluated through the VABS, have been confirmed in different studies (Kanne et al., 2011; Klin et al., 2007; McDonald et al., 2015; Perry et al., 2009; Saulnier & Klin, 2007). In particular, adaptive behavior impairments are more evident in people with low-functioning ASD (LFASD), in which adaptive behavior appears consistent with cognitive level. However, these impairments are also present among people with high-functioning ASD (HFASD), in which there is a gap between IQ and adaptive functioning that increases with age (Kanne et al., 2011; Klin et al., 2007; Liss et al., 2001; Matthews et al., 2015; Szatmari et al., 2009). Conversely, Liss et al. (2001) suggest that, while IQ is predictive of adaptive behavior in people with LFASD, other characteristics in people with HFASD, such as autism symptoms, language and verbal memory, affect adaptive behavior, and may explain the presence of a gap in people with HFASD.

Some studies have documented a negative correlation between age and adaptive functioning (Kanne et al., 2011; Klin et al., 2007; Perry et al., 2009; Saulnier & Klin, 2007). However, this trend conflicts with longitudinal studies,

which suggest that greater autonomy in daily life skills as people age (Green & Carter, 2014; Smith et al., 2012).

Studies using the ABAS have indicated that there is no relationship between either age or IQ and adaptive behavior (Kenworthy et al., 2010; Lopata et al., 2012). In particular, Kenworthy et al. (2010) found a positive relation only between IQ and the Conceptual domain of the ABAS-II in HFASD adolescents aged from 12 to 22 years. However, they did not find a significant relationship between age and adaptive skills. The same results are confirmed by Lopata et al. (2012). It is important to note that these studies, which are among the few to use ABAS-II scales, both utilized homogeneous HFASD samples. More recently, McDonald et al. (2016) investigated adaptive functioning in two groups aged 7–12 with HFASD and LFASD. In contrast with results found by Kenworthy et al. (2010) and Lopata et al. (2012), their study found that in children with HFASD, adaptive skills were significantly related to age whereas no relationship was found with IQ. In children with LFASD, no significant relationship between adaptive functioning and age was found, but IQ was significantly related to overall adaptive ability and all domains of the ABAS-II. These findings appear to suggest that as cognitive ability increases, the adaptive skills of a child with LFASD are also expected to increase.

More mixed results have been obtained when the relationship between adaptive functioning and ASD symptoms was analyzed (Lopata et al., 2012; McDonald et al., 2015), especially when the VABS had been used. While some studies reported a negative correlation between adaptive behavior and ASD symptoms (Liss et al., 2001; Perry et al., 2009), others did not confirm this relationship, suggesting that adaptive behaviors could be relatively independent of ASD clinical symptoms (Kanne et al., 2011; Klin et al., 2007; Saulnier & Klin, 2007). However, the results are homogeneous in studies where the ABAS-II was used, confirming a negative relationship between adaptive functioning and ASD symptoms (Kenworthy et al., 2010; Lopata et al., 2012). In particular, Kenworthy et al. (2010) found that the ABAS-II overall adaptive functioning (e.g., General Adaptive Composite) and domains are negatively associated with the scores derived from the Autistic Diagnostic Observation Schedule (ADOS; Lord et al., 1999), addressing communication and social interaction areas. No relationship emerged between ADOS stereotyped behavior and ABAS-II scores. Different results were found when the Autism Diagnostic Interview-Revised (ADI-R; Le Couteur et al., 2003) was used. In the study carried out by Lopata et al. (2012), the total score on the ABAS-II are negatively associated with the total score on ADI-R. Moreover, a negative relationship was found between the area of restricted, repetitive, and stereotyped patterns of behavior, derived from the ADI-R and ABAS-II social and practical domains.

These mixed results are probably due to variations in the different tools used to assess ASD symptoms in different studies, such as the ADOS (Lord et al., 2000), ADI-R (Rutter et al., 2003), the Wing Autism Diagnostic Interview Checklist (WADIC; Rapin, 1996) and the Childhood Autism Rating Scale (CARS; Kanne et al., 2011; Kenworthy et al., 2010; Klin et al., 2007; Liss et al., 2001; Lopata et al., 2012; Perry et al., 2009; Saulnier & Klin, 2007; Schopler et al., 1988). Nevertheless, in most of these studies homogeneous samples HFASD were enrolled. This issue might contribute to the mixed findings among studies (Lopata et al., 2012; McDonald et al., 2016). Klin et al. (2005) observed that the typical VABS profile for ASD individuals is characterized by weaknesses in socialization and, although to a lesser extent, communication. On the other hand, there are strengths in the skills needed for everyday life (e.g. personal care and care living in the community). McDonald et al. (2015) confirmed this results in a sample of children with HFASD.

Beyond the VABS/VABS-2, little research has been carried out using the ABAS-II, although it is the only adaptive functioning measure developed according to the definition of the American Association on Intellectual and Developmental Disabilities (AAIDD,), addressing three broad domains of adaptive behavior: Conceptual, Social and Practical (Kenworthy et al., 2010; Lopata et al., 2012; McDonald et al., 2015).

Kenworthy et al. (2010) examined the adaptive functioning profiles of adolescents with HFASD, as assessed by the ABAS-II. They found a general impairment in all three ABAS-II domains. Specifically, Social and Home living scales arose as weaknesses, whereas the Functional Academics scale scores showed a strength. Similar results emerged in Lopata et al. (2012), examining the ABAS-II adaptive functioning profile in youth with HFASD aged 7–12 years. Specifically, they found significant deficits on General Adaptive Composite and all three adaptive composites, with the lowest scores in Social, Home living and Self-Direction. In a subsequent study, Lopata et al. (2013) compared the profile yielded from the VABS-II, the ABAS-II with the BASC-II (Reynolds & Kamphaus, 2004) parent-rated adaptive behavior of 6- to 11-year-olds with HFASD. The three adaptive measures revealed significant deficits, with the VABS-II and the ABAS-II indicating relative weaknesses in social skills and strengths in academic-related skills.

In study by McDonald et al. (2016) comparing children aged 6–12 years with HFASD and LFASD, the authors showed that the first group had cognitive strengths and adaptive deficits, while children with LFASD had better adaptive skills than cognitive abilities. This contrasts with studies using VABS-II that suggest a congruence between cognitive level and adaptive functioning (Kanne et al., 2011; Klin et al., 2007).

The Object of the Current Study

As a whole, the evaluation of adaptive behavior is far from easy, especially in people with low skill levels, who are more likely to be subject to a floor effect. Consequently, it is very common to find flat profiles that emphasize weaknesses and obscure strengths. The current study aims to counteract the floor effect in the assessment of adaptive behavior of adolescents with ASD, as derived by the Italian version of ABAS-II (Ferri et al., 2014). In reaching this goal, the statistical Hessel method (Hessel et al., 2009) and its revision by Orsini et al. (2015) were used.

Method

Participants and Procedure

The group included 47 mothers (Mean Age = 54.61; SD = 4.63) of adolescents (13 girls and 34 boys) with ASD who were attending public middle and high schools in Rome. The adolescents ranged in age from 12 to 18 years (Mean Age = 15.87; SD = 1.68) and met DSM-IV-TR criteria for ASD (Autism or Pervasive Developmental Disorder Not Otherwise Specified), confirmed prior to their participation in the study by experienced psychiatrists and psychologists (IQ total score, $M = 63.57$, $SD = 14.02$). Among mothers, 38.3% ($n = 18$) had attended high school, 46.81% university ($n = 22$), and 14.89% ($n = 7$) had completed a PhD course or other advanced specialization.

Caregivers, who provided informed consent, were asked to complete the Italian version of the ABAS-II (Ferri et al., 2014) to assess the adaptive functioning of their daughters and sons. The administration took approximately 30 min to complete. To ensure higher reliability, the Italian version of ABAS-II was administered to each participant's parent by the same professional.

This study was reviewed and approved by the Ethics Commission of XXX (*Blinded for review*). For clarity, we will refer to the adolescents themselves as “participants” throughout this paper, although their mothers completed the measures.

Instruments

The Italian version of the Adaptive Behavior Assessment System—Second Edition (ABAS-II) Parent Form (5–21 years; Ferri et al., 2014) is a comprehensive measure of adaptive functioning at home and within the community. It consists of 232 items, each rated on a scale from 0 (not able) to 3 (always/almost always).

The parental form includes 10 skill areas—Communication, Community Use, Functional Academics, Home Living,

Health and Safety, Leisure, Self-Care, Self-Direction, Social, and Work [Work is only completed for individuals ≥ 17]—which are combined to form three composite indices, including Conceptual (CON; Communication, Functional Academics, Self-Direction), Social (SOC; Leisure, Social), and Practical (PRAC; Self-Care, Home Living, Community Use, and Health and Safety), as well as an overall General Adaptive Composite (GAC). The ABAS-II composite scores have a mean of 100 ($SD = 15$), while scores for each skill area have means of 10 ($SD = 3$). Higher scores represent better adaptive functioning. For details about the psychometric properties of the Italian version of the ABAS-II, see Ferri et al. (2014).

Data Analysis

Step 1: Canonical Transformation of Raw Scores

Raw scores on the ABAS-II subtests were converted into weighted scores (mean = 10 and standard deviation = 3, with a range from 1 to 19) according to the conversion tables of the Italian ABAS-II standardization. Domain and composite scores have a norm-referenced $M = 100$ and $SD = 15$. In the Italian ABAS-II standardization sample, the lower adaptive domain scores are on Conceptual Index (57), Social Index (50), Practical Index (44) and General Adaptive Composite (40). Then, the percentage of adolescents with ASD who achieved a weighted score of 1 on nine subtests was calculated.

Step 2: New Transformation as Proposed by Hessel et al. (2009) and Orsini et al. (2015)

Using the method proposed by Hessel et al. (2009) and extended by Orsini et al. (2015), the raw scores of subtests that correspond to a weighted score (ws) of 1 were transformed into z -scores using means and standard deviations of the raw scores of the subtests for each age group. All other weighted scores higher than 1 were transformed into z -scores using the inverse formula of their composition in the standardization process: $z = (ws - 10)/3$. The sums of the z -score components of the three indices (Conceptual, Social, Practical) and the General Adaptive Composite Index, were calculated, transforming these sums into new z -scores and later into standard scores with a mean of 100 and a standard deviation of 15. In this way, the resulting measures of indices are expressed in z -scores, and therefore, the floor effect presented by a minimum score of 40 is overcome. For more detailed information about this statistical method, see Orsini et al. (2015) and Pezzuti et al. (2018). In summary, in order to overcome the floor effect, while Hessel's method explains how to convert the raw scores into the weighted

scores, Orsini's method goes a step further in calculating the indices.

Consider, for example, the raw scores of an 18-year-old male on the three subtests (Communication, Functional Academics and Self-Direction) of the Conceptual Index, reported in column A of Table 1. Using the canonical method, such raw scores would be all converted to a weighted point of 1 (see column b) corresponding to a normalized z -score of -3 (see column c), corresponding (after appropriate psychometric transformations) to an adaptive domain score of 57. If instead one uses the means and standard deviations of the age-relevant reference sample of the subject (see column d), the new z -scores are lower (see column e). Then, the z -scores (column f) are converted to a new z -score by the means and standard deviations of the weighted scores of the entire standardization sample (column g) and finally converted to adaptive domain score (column h) and finally converted to adaptive domain score ($M = 100$; $SD = 15$) by the formula $Z(15) + 100$ (see column i) to obtain the new Conceptual Index. We can see that the new Conceptual Index value drops to 10 points, which is lower than the 57 points obtained using the canonical method.

Results

Step 1: Analysis of Canonical Scores on the ABAS-II for the ASD Sample

Using the canonical transformation of the raw scores according to the conversion tables of the ABAS-II Italian standardization (Ferri et al., 2014), a very high percentage of subtests with a weighted score of 1 occurred in the sample. Table 2 shows the percentage of adolescents with ASD who achieved a weighted score of 1 on nine of the ABAS-II subtests. In detail, there were subtests that, more than others, had a weighted score of 1, such as Social, (74.5%), followed by Functional Academics (61.7%), Communication (59.6%), Community Use and Self-Direction (both 57.4%), and Leisure (55.3%). So, the three subtests of Conceptual and the two subtests of Social each had a higher frequency of weighted scores of 1. The smallest proportion occurred in Home Living (25.5%).

There was a total of 423 scores on the subtests (47 subjects $\times$ 9 subtests); 220 scores (52%) correspond to a weighted score of 1 (Table 3). This data confirmed the level of impairments in adolescents with ASD.

Table 4 shows the means and standard deviations of weighted scores (ws) for the nine ABAS-II subtests and the values expressed into z -scores using the inverse formula of their transformation: $z = (ws - 10)/3$. These means are 2 to 2.5 standard deviations below the mean. This transformation in z -scores is necessary to compare them with the

Table 1 Example of computing new conceptual index on data of one boy of 18 years of age

Subtest	A	B	C	d	e	f	g	h	i
	Raw score (x)	Range of raw score corresponding to a weighted score (ws) of 1	Normalized x scores = $(ws - 10)/3$	Mean (M) and standard deviation (sd) of the raw scores of the subtests on group of 18 years of age	z new points (z_n) with Orsini et al. (2015) method: $z = (x - M)/sd$	Sum of the z_n points about Conceptual Index	Mean (M_w) and standard deviation (sd_w) of weighted scores about Conceptual Index (in Italian standardization)	$Z = (z_n - M_w)/sd_w$	$IQ_{Cone} = Z(15) + 100$
Communication	32	0–50	–3	65.83 (5.31)	–6.37	–14.47	–0.03 (2.4)	–6.02	10
Functional academics	35	0–46	–3	62.31 (5.94)	–4.60				
Self-direction	34	0–40	–3	63.43 (8.40)	–3.50				

Table 2 Percentage of adolescents with ASD with weighted score of 1 to 9 core ABAS-II subtest

Subtest	F	f%
Communication	28	59.6
Community use	27	57.4
Functional academics	29	61.7
Home living	12	25.5
Health and safety	18	38.3
Leisure	26	55.3
Self-care	18	38.3
Self-direction	27	57.4
Social	35	74.5

Table 3 Percentage and cumulative (cum) percentage of weighted scores in the sample

Weighted score	Number of subtests	Percentage	Cum %
1	220	52.0	52.0
2	18	4.3	56.3
3	20	4.7	61.0
4	18	4.3	65.2
5	26	6.1	71.4
6	21	5.0	76.4
7	11	2.6	79.0
8	24	5.7	84.6
9	20	4.7	89.4
10	19	4.5	93.9
11	4	0.9	94.8
12	8	1.9	96.7
13	5	1.2	97.9
14	5	1.2	99.1
15	2	0.5	99.5
16	1	0.2	99.8
17	1	0.2	100.0

47 adolescents $\times$ 9 subtests = 423 subtest scores

results of the method proposed by Hessler et al. (2009) and Orsini et al. (2015), which were shown in Step 2.

The means of the three indices computed using the canonical method for our participants were 66.81 (SD = 17.57) for Conceptual, 58.55 (SD = 14.16) for Social, and 64.45 (SD = 19.90) for Practical, with a range between averages of 5.5 points and with Social emerging as lower than the other two indices. When the pairs of indices were compared, ANOVAs showed significant differences only for Conceptual/Social ($F_{(1,92)} = 6.29$; $p = 0.01$; $\eta^2 = 0.06$). In summary, a fairly flat pattern of scores was observed.

Table 4 Means and standard deviations (SDs) of the weighted scores of nine subtests calculated according to traditional method and their transformation into z points

Subtest	Weighted scores		Normalized z scores	
	M	SD	M	SD
Communication	3.62	3.75	−2.13	1.25
Community use	3.77	4.16	−2.08	1.39
Functional academics	4.30	4.95	−1.90	1.65
Home living	5.36	3.81	−1.55	1.27
Health and safety	4.79	3.83	−1.74	1.28
Leisure	2.66	2.60	−2.45	0.87
Self-care	4.19	3.50	−1.94	1.17
Self-direction	3.40	3.23	−2.20	1.08
Social	2.40	2.95	−2.53	0.98

Step 2: Use of Raw Scores with New Method and Comparison with the Canonical Method

A second transformation of the raw scores was carried out using the method proposed by Hessler et al. (2009) and extended by Orsini et al. (2015) to allow the calculation of indices, as described in the Method section. The steps are shown in Table 1 from column F to column I. Table 5 shows means and standard deviation of the z-scores of the nine ABAS-II subtests calculated using the two methods.

Compared with the canonical method, the method developed by Orsini et al. (2015) allowed a greater decrease in means, a significant increase in standard deviations and essentially a much larger interindividual variability. A similar result emerged for the indices (Table 6), where a wider variability of scores is obtained with Orsini et al.'s (2015) method compared to the traditional method.

As shown in Table 7, the new adaptive domain scores are lower than the means calculated with the traditional method, however the differences between the new pairs of means are not statistically significant (see Table 8). Analysis of pairs of subtests on each index were performed, resulting in the following (see Table 8): In the pair comparison between Community Use, Home Living, Health and Safety and Self-Care of participants there were more weaknesses in Self-Care. In the comparison between Community Use and Home Living there were more weaknesses in Community Use. In the comparison between Leisure and Social there were more weaknesses in Social.

Finally, when we analyzed the Adaptive Domain Index calculated with the new method, they ranged between 1 and 112, but there are 14 adaptive domain scores (30%) off the scale, i.e. with $z < -6.66$ points (i.e. 14 Adaptive Domain Index scores are below 1).

Table 5 Means and standard deviation (SD) of z points to subtests calculated with the traditional method and that developed by Orsini et al. (2015) methods

Subtest	Normalized z scores		Z points with the Orsini et al. (2015) method		$F_{(1,92)}$	p	Partial η^2
	M	SD	M	SD			
Communication	−2.13	1.25	−4.10	3.56	12.81	<0.001	0.12
Community use	−2.08	1.39	−3.17	2.50	06.82	<0.01	0.07
Functional academics	−1.90	1.65	−4.95	4.59	18.38	<0.001	0.17
Home living	−1.55	1.27	−1.66	1.47	00.16	<0.680	0.00
Health and safety	−1.74	1.28	−2.79	3.02	04.90	<0.030	0.05
Leisure	−2.45	0.87	−3.52	2.04	10.89	<0.001	0.11
Self-care	−1.94	1.17	−6.22	8.44	11.85	<0.001	0.11
Self-direction	−2.20	1.08	−3.31	2.27	09.22	<0.001	0.09
Social	−2.53	0.98	−5.44	3.70	27.28	<0.001	0.22

Partial eta-squared values were calculated as a measure of effect size, and results were interpreted using Cohen's (1988) guidelines for determining small (0.01), medium (0.06), and large (0.14) effects

Table 6 Means and standard deviation of z points to indices calculated with the traditional and the Orsini et al.'s (2015) methods respectively

Indices	Normalized z scores		Z points with the Orsini et al.'s (2015) method		$F_{(1,92)}$	p	Partial η^2
	M	SD	M	SD			
Conceptual	−6.23	3.60	−12.37	09.96	15.80	<0.001	0.15
Social	−4.98	1.69	−8.96	05.55	22.11	<0.001	0.19
Practical	−7.30	4.50	−13.84	14.26	08.98	<0.001	0.09

Partial eta-squared values were calculated as a measure of effect size, and results were interpreted using Cohen's (1988) guidelines for determining small (0.01), medium (0.06), and large (0.14) effects

Table 7 Means and standard deviation of adaptive domain scores to indices calculated with the traditional and the Orsini et al.'s (2015) methods respectively

Indices	Traditional method				Method of Orsini et al. (2015)				$F_{(1,92)}$	p	Partial η^2
	Min	Max	M	SD	Min	Max	M	SD			
Conceptual	57.0	117.0	66.81	17.56	−96.5	116.9	22.92	62.23	21.65	<0.001	0.19
Social	50.0	100.0	58.55	14.16	−58.9	100.0	23.90	47.16	23.29	<0.001	0.20
Practical	44.0	111.0	64.45	19.89	−193.3	118.3	29.47	73.78	09.85	<0.001	0.09

Partial eta-squared values were calculated as a measure of effect size, and results were interpreted using Cohen's (1988) guidelines for determining small (0.01), medium (0.06), and large (0.14) effects

Discussion and Conclusion

ASD is a long-life disorder characterized by social and communication deficits with narrow and repetitive behaviors and interests (APA, 2013). The assessment of adaptive behavior is central to identify the severity of impairments in self-sufficiency, independence and daily functioning (Klin et al., 2005) as well as to plan goals in clinical, family and school settings (Tasse et al., 2012). However, the evaluation of adaptive behavior may be challenging with people who have lower levels of abilities. The main problem is linked to the floor effect, which produces flat

profiles. Consequently, strengths are often concealed, and weaknesses appear predominant.

The aim of this study is to counteract floor effects in the evaluation of adaptive behavior in adolescents with ASD, as derived from the Italian version of the ABAS-II (Ferri et al., 2014). Specifically, we compared the profiles yielded by the traditional method—transformation of raw scores into weighted scores—with those computed through the statistical Hessl method (Hessl et al., 2009) and its extension developed by Orsini et al. (2015). Previous studies (Orsini et al., 2015; Pezzuti et al., 2018) demonstrated

Table 8 Comparisons between pairs of subtests and indices using the Orsini et al.'s (2015) method

Adaptive domain indices	Comparison between subtest (in z means points)	M	SD	M	SD	F	p	Partial Eta ²
		Subtest 1		Subtest 2				
Conceptual	Communication vs. Functional academics	−4.10	3.56	−4.95	4.59	1.01	0.310	0.01
	Communication vs. Self-direction	−4.10	3.56	−3.31	2.27	1.63	0.200	0.02
	Functional academics vs. Self-direction	−4.95	4.59	−3.31	2.27	4.82	0.030	0.05
Social	Leisure vs. Social	−3.52	2.04	−5.44	3.70	9.81	<0.001	0.09
Practical	Community use vs. Home living	−3.17	2.50	−1.66	1.47	12.70	<0.001	0.12
	Community use vs. Health and safety	−3.17	2.50	−2.79	3.02	0.42	0.520	0.01
	Community use vs. Self-care	−3.17	2.50	−6.22	8.44	5.64	0.020	0.06
	Home living vs. Health and safety	−1.66	1.47	−2.79	3.02	5.37	0.020	0.05
	Home living vs. Self-care	−1.66	1.47	−6.22	8.44	13.28	<0.001	0.13
	Health and safety vs. Self-care	−2.79	3.02	−6.22	8.44	6.84	<0.010	0.07
Adaptive Domain Indices	Comparison between indices (in standardized points)	Index 1		Index 2				
	Conceptual vs. Social	22.92	62.23	23.90	47.16	0.01	0.930	0.00
	Conceptual vs. Practical	22.92	62.23	29.47	73.78	0.21	0.640	0.00
	Social vs. Practical	23.90	47.16	29.47	73.78	0.19	0.660	0.00

Partial eta² values were calculated as a measure of effect size, and results were interpreted using Cohen's (1988) guidelines for determining small (0.01), medium (0.06), and large (0.14) effects

that the latter method overcomes the limits underpinning floor effect problems through z-score normalization.

As hypothesized, a higher variation in adaptive behavior in adolescents with ASD was found with Orsini et al.'s (2015) method compared to the traditional method. In particular, there was a general impairment across all three domains (Conceptual, Social, and Practical domains) of the ABAS-II and there is no greater impairment of Social domain than the other two—Conceptual and Practical. Moreover, within the Social domain, we found a greater impairment in Social than Leisure skills. These results are consistent with study by Kenworthy et al. (2010) and they can be explained if we analyze the items of Leisure and Social subscales. The Social subscale is characterized by items that assess purely social skills, such as the ability to relate to others, awareness of their own and other people's emotions, friendship, respect for shared social rules, courtesy skills (e.g. saying “thank you,” “please,” “sorry”). The items of the Leisure subscale, instead, involve questions that concern many skills and activities that are “non-social,” such as playing solitaire, listening to music, reading a book, organization skills and free time planning (Kenworthy et al., 2010). Thus, it could be hypothesized that the inclusion of these “non-social” elements in the Social composite score leads to an underestimation of social deficits in ASD (Kenworthy et al., 2010).

The results concerning the Conceptual and Practical domains conflict with evidence provided by the literature (Kenworthy et al., 2010; Lopata et al., 2012). Regarding the Conceptual domain, our results seem to suggest

a greater weakness in Functional Academics, in contrast with previous studies that found both a strength in these abilities together and a positive correlation between IQ levels and Conceptual scores (Kenworthy et al., 2010; Lopata et al., 2012). These results could be explained by focusing on IQ levels. In studies carried out by Kenworthy et al. (2010) and Lopata et al. (2012), the HFASD participants had IQs > 100, whereas in our study, the LFASD adolescents were enrolled.

When the subscales of the Practical domain were analyzed, we found a greater impairment of Self-Care skills. Conversely, Home Living skills appeared as a strength compared to other skills. These results disagree with Kenworthy et al. (2010), who suggested a greater weakness in Home living than Community Use, Self-Care and Health and Safety abilities. This may be due to a cultural difference between American and Italian families, but further research would be needed to explore the role of cultural differences in the development of adaptive behavior.

During adolescence, adult-mediated social relationships tend to diminish compared to childhood and the need for adolescents to be able to relate to others independently increases. The typical characteristics of ASD make the development of this social autonomy even more complicated, so it is essential to design interventions aimed at maximizing the opportunities for socialization. By increasing social opportunities, the adolescent with ASD will have more opportunities to engage with peers and this engagement may be helpful in improving other aspects related to autonomy in daily life.

As a whole, the current study confirms the goodness of the procedure developed by Hessel et al. (2009) and revised by Orsini et al. (2015) to overcome floor effect problems in the evaluation of adaptive behaviors in adolescents with ASD, especially those with LFASD. In light of this issue, we recommend the use of renormalized scores in both research and clinical areas for its practical implications. From a research perspective, the renormalization and improved sensitivity of adaptive behavior testing for individuals with ASD, especially those with low functioning, may allow for more powerful research designs, contributing to further understanding of the interplay between behavioral, biological and cognitive phenotypes. From a clinical point of view, the renormalized scores can provide a more accurate profile of strengths and weaknesses, allowing clinicians and educators to plan intervention programs and monitor progress over time.

Despite these appreciable results and implications, this study has some limitations. First, the low number of participants in the current study necessitates caution in interpreting the results. Second, our assessment of adaptive behavior is limited to the use of the ABAS-II. This is a reliable and valid tool; however, the comparison between performances obtained by different instruments may be informative regarding the validity of this approach to overcome flooring problems. Moreover, a restricted age range was considered, although the ABAS allows to evaluate adaptive behavior from childhood to adulthood. Future research could expand the age range or consider different age groups to study any differences in adaptive skills of the population with ASD.

Finally, all of our participants were adolescents with ASD, future investigations should examine the sensitivity of adaptive behavior testing in other neurodevelopmental disorders at different ages.

Authors Contributions The authors discussed the contents of this article together. Fiorenzo Laghi, Rosa Ferri and Lina Pezzuti conceptualized the study elaborated on the research design and hypotheses, and contributed to the interpretation of research findings. Elvira Caramuscio, Antonia Lonigro, and Monica Rea collected data and contributed to writing and editing of this paper and collected the references and carried out management activities to scrub and maintain research data. Lina Pezzuti and Fiorenzo Laghi devised the methodological content, analyzed the data, and contributed to the interpretation of research findings. Fiorenzo Laghi, Rosa Ferri, and Lina Pezzuti wrote the final version of this paper.

Declarations

Conflict of interest We confirm that we have not conflict of interest in this research.

Ethical Approval The research has been approved by the ethical committee of the Department of Dynamic and Clinical Psychology, and

Health Studies of the Faculty of Medicine and Psychology of the Sapienza University of Rome.

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